Module 7: Classification

Part 1: Logistic Regression

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Some lecture slides and instructional materials in this course are adapted from the following sources:

- *[1] https://blog.alliedoffsets.com/understanding-the-difference-between-linear-and-logistic-regression-models*

Example of classification problems

- Goal: carry out classification of a $\underbrace{\text{response } Y}_{\text{categorical}}$ using predictors $X_1, \ldots, X_p$.

- Main difference with the regression models we've covered so far is that now Y is qualitative (categorical).

- Still in the supervised learning setting. Classification is NOT clustering!!
  ↳ inputs (x) are labeled (y)

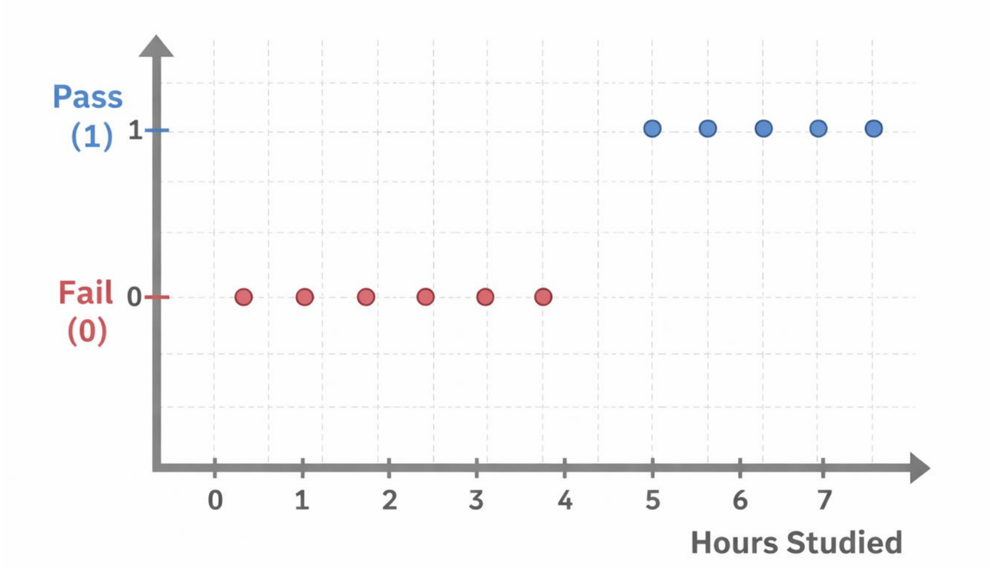
Example of classification problems

- Email is spam or not spam?
- Is this transaction a fraud or not fraud?
- Does an individual have a disease or not a disease? Is this DNA mutation harmful or not?
- Is this image of a dog or not a dog (image classification)?
- ....

Example

Suppose we are trying to classify the test result of students:

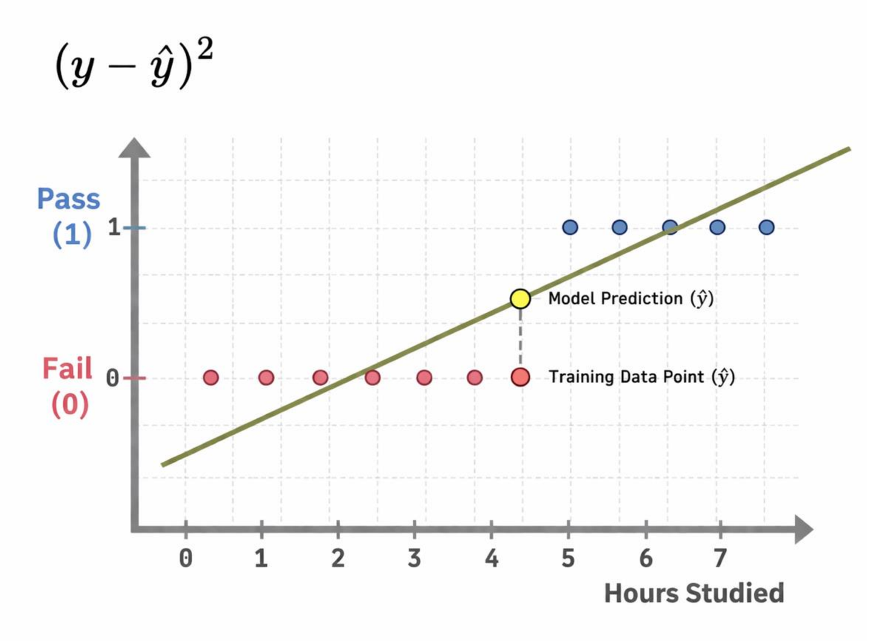
$$Y = \begin{cases} 1 & \text{if } pass \\ 0 & \text{if } fail \end{cases}$$



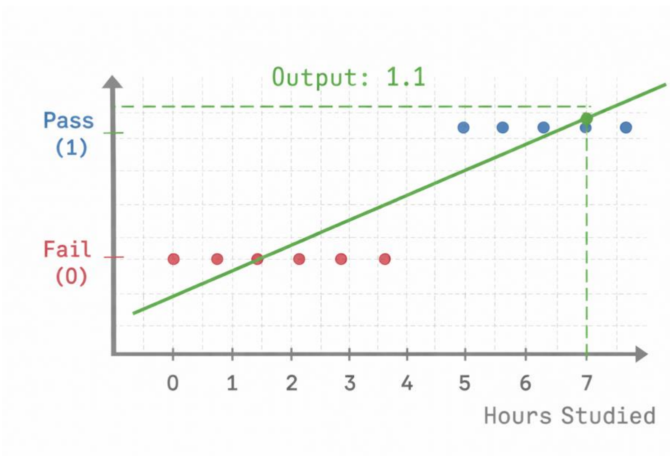
Example

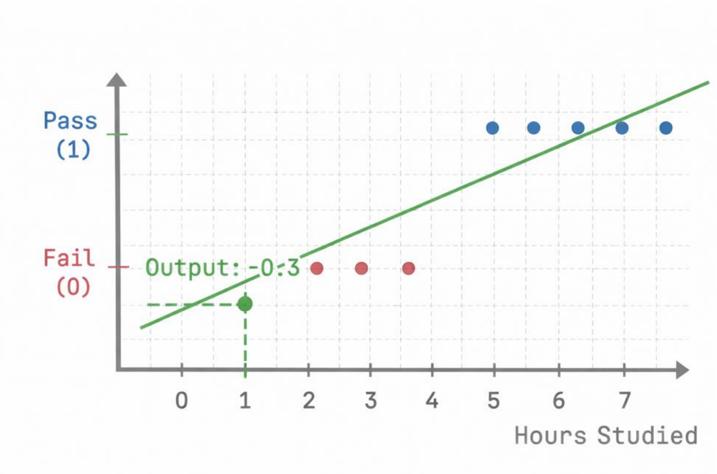
Why don't we use a least squares model here to predict Y ?

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 * number\ of\ hours\ studied$$



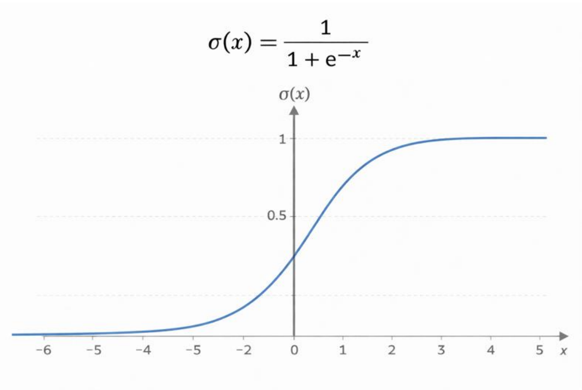
Example





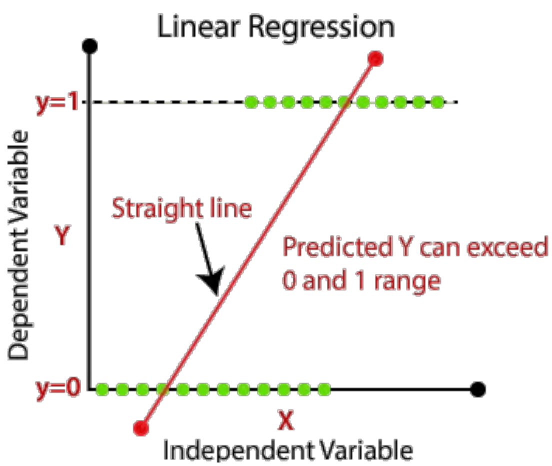
- In classification, we usually want something interpretable as a probability (between 0 and 1).
- Linear regression can give values **< 0 or > 1**, which don't make sense as probabilities.

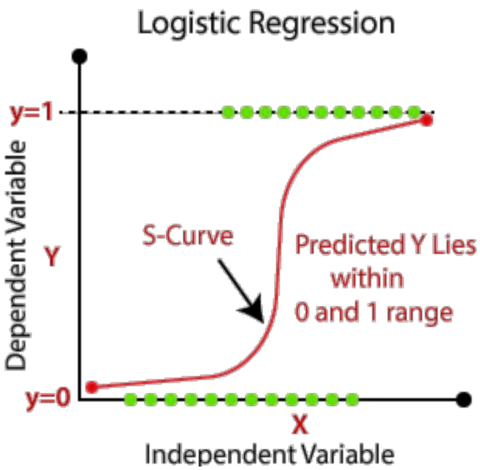
Sigmoid function



→ always out put values between 0 and 1

→ S-curve





Estimating beta

Instead we need to use classification techniques that are specifically designed to handle this kind of problem:

1. **Logistic regression**
2. Linear/quadratic discriminant analysis
3. Naive Bayes
4. k-nearest neighbor

Gold standard for classification is **Bayes Rule**.

Bayes Rule

Let's start with a Y that only takes 2 values:

$$Y = \begin{cases} 1 & \text{if } \mathit{pass} \\ 0 & \text{if } \mathit{fail} \end{cases}$$

Target: $P\left(Y = 1 \mid X_1, \ldots, X_p\right)$.

If I knew this probability, I could use Bayes Rule directly:

$$\begin{cases} P(Y = 1 \mid X) > 0.5 & \text{then } \hat{Y} = 1 \\ P(Y = 1 \mid X) \leq 0.5 & \text{then } \hat{Y} = 0 \end{cases}$$

Logistic Regression

Instead of modeling Y directly, let's try to model the probability that Y belongs to a category.
Suppose Y only takes 2 values:

$$Y = \begin{cases} 1 & \text{if } pass \\ 0 & \text{if } fail \end{cases}$$

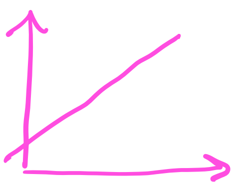
Target: $P\left(Y = 1 \mid X_1, \ldots, X_p\right) = p(x)$.

- $p(x)$ is usually unknown to us, so we need to estimate it from the data.
- We must model $p(x)$ using a function that can guarantee outputs fall between 0 and 1 .

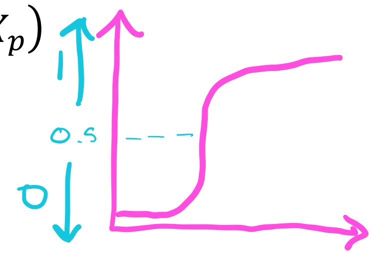
In logistic regression, we use the logistic function:

Least Squares:

$$p(x) = \frac{\exp\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p\right)}{1 + \exp\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p\right)}$$



$$\sigma(x) = \frac{1}{1+e^{-x}} \Rightarrow \frac{e^x}{1+e^x}$$



$$p(x) = \frac{\exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}{}$$

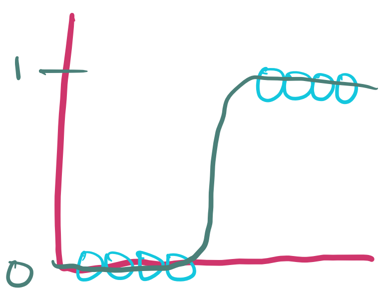
$$\log\left(\frac{p(x)}{1 - p(x)}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Estimating β

$\beta_0, \beta_1, \dots, \beta_p$ are parameters. These are unknown quantities and we need to estimate them from our data $\Rightarrow$ **Maximum likelihood estimation.** (MLE)

- Intuition: We seek estimate for $\beta_0, \beta_1, \dots, \beta_p$ such that the predicted probability $\hat{p}(x) = \hat{P}(Y = 1 \mid X)$ is large for observations where $Y = 1$ and small for observations where $Y = 0$.
- Following our example, that means
- If an individual student passed the exam($Y = 1$), then we want $\hat{p}(x)$ to be close to 1 .
- If an individual student failed the exam ($Y = 0$), then we want $\hat{p}(x)$ to be close to 0 .

prob of passing



$Y=1$, $P(X)=1$ → if the student passed, we want the prob to be close to one

Maximum likelihood estimation technical details (sort of)

① student 1 = pass, $y_1 = 1$, P_1 (we want $P_1 \to 1$)

② student 2 = fail, $y_2 = 0$, P_2 (we want $\underbrace{P_2 \to 0}$)

$\Rightarrow 1 - P_2 = 1$

$L = P_1^{y_1}(1-P_1)^{1-y_1} \cdot P_2^{y_2}(1-P_2)^{1-y_2} \to$ Likelyhood

Maximize Likelyhood = what values make this large
($L = 1$)

① student 1, $P_1 \approx 1$, $P_1^1(1-P_1)^{1-1} = 1 \cdot 1 = 1$ $\quad 1 \times 1 = 1$ (max)

② student 2, $P_2 \approx 0$, $P_2^0(1-P_2)^{1-0} = 1 \cdot 1 = 1$

$L(B_0, B_1, \ldots, B_p) = \prod_{i=1}^{n} f(y_i = p)$ ← product

} no closed form
gradient descent model

R code

See R script: logit_intro.R